

# HARIS WAHEED BHATTI

BIM Specialist | Information Manager (ISO 19650)

Ljubljana, Slovenia | +386 69 437 910 | harisw95@gmail.com

linkedin.com/in/bhatti25 | github.com/alfarseniko

Open to relocating to the Netherlands; available immediately.

## PROFILE

Civil engineer with a European Master's in Building Information Modelling and Bachelor's coursework in hydrology, water resources and geographic information systems (AutoCAD, HEC-HMS, HEC-RAS, ArcGIS). Comfortable combining fieldwork with technical documentation, having coordinated on-site design refinement and monitored progress and quality during an industrial engineering internship. Strong written and spoken English (C1).

## PROJECTS

**Federated BIM Model – Pakistani-Themed Expo, Guimarães** | IFC, Autodesk Construction Cloud ..... Oct 2025

- Owned the federated model and clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650.

**ifctoneo4j – Python package (PyPI)** | Python, IfcOpenShell, Neo4j, Cypher ..... 2026

- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published to PyPI: `pip install ifctoneo4j`.

**BIMGraphExplorer – LLM agent for IFC question answering** | Python, MCP, Neo4j ..... 2026

- Adaptive querying system exposing four read-only tools over the Model Context Protocol; profiles the graph schema and issues iterative Cypher queries conditioned on prior results.
- Raised answer accuracy by 33 percentage points and source transparency from 12.5% to 99.7% on a fixed model and dataset, through prompt design alone.
- Benchmarked against a published direct-IFC baseline under matched conditions (same model, same validated judge; 23 IFC datasets, 367 questions, 15 projects). Overall accuracy 72.0% against the baseline's 82.3%, but reached source-transparency parity (99.7% versus 100%) and outperformed it on 5 of 23 datasets, by up to 24 points on one.

## EDUCATION

**MSc, European Master in BIM (BIM A+)** | UL FGG, Ljubljana, Slovenia ..... Oct 2025 – Jul 2026

- Field of study: Information Management in Construction. Completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade 9.27/10.
- Thesis: *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. [hdl.handle.net/20.500.12556/RUL-185169](https://hdl.handle.net/20.500.12556/RUL-185169)
- Applied coursework spanning the BIM delivery chain: IFC schema auditing and model remediation, parametric façade optimisation in Dynamo, process modelling and workflow improvement, 5D quantity take-off and cost estimation, and scan-to-BIM from point-cloud capture.

**BE Civil Engineering** | NUST, Islamabad, Pakistan. Final grade 3.25/4.00 ..... Sep 2020 – Jun 2024

- Coursework in hydrology and water resources engineering, including hydraulic modelling with AutoCAD, HEC-HMS and HEC-RAS.

## EXPERIENCE

**Civil Engineering Intern** | Fatima Fertilizer Company, Pakistan ..... July 2023

- Modelled a 6,804 m<sup>2</sup> rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement.

**Contract Management Intern** | Engineering and Management Services, Pakistan ..... June 2023

- Reviewed contract reports and arbitration documentation under FIDIC conditions; presented findings to the internal legal team.

**Research Intern (remote)** | Southern Illinois University Edwardsville, USA ..... July 2023

- Synthesised academic literature to support a faculty-mentored research proposal.

## SKILLS

**BIM & AEC:** AutoCAD, ArcGIS, HEC-HMS, HEC-RAS, IFC (IFC2X3, IFC4, IFC4X3), ISO 19650, Employer's Information Requirements, clash detection, federated models, Revit, Dynamo, Autodesk Construction Cloud

**Programming & Data:** Python, TypeScript, Neo4j, Cypher, knowledge graphs, Git

**AI / LLM:** Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking

**Languages:** English (C1)